Quantifying CNN Robustness to Weather Corruptions in Traffic Sign Recognition

Course: Stats 426 | **Team:** Ayah Halabi, Setara Nusratty, Mark Rahal, Hochan Son | **Timeline:** 4 weeks

Objective

Deep CNNs achieve 99% accuracy on clean traffic sign datasets but degrade significantly under real-world conditions. We quantify this robustness gap by measuring ResNet-50's performance on synthetic weather corruptions (rain, fog, low brightness) to identify which conditions and sign types are most vulnerable.

Methodology

Dataset: GTSRB (German Traffic Signs) and LISA (US Traffic Signs)

- **Scope:** 5 classes (Stop, Yield, Speed 30/50/70 km/h)
- **Resolution:** 224×224 RGB images
- **Split:** 80% train / 20% validation (from training set) + separate test set

Model: ResNet-50 (pretrained on ImageNet)

- Fine-tune on clean 5-class subset
- Standard hyperparameters: Adam (lr=1e-4), 15 epochs, early stopping
- Target: >90% clean validation accuracy

Corruptions (Albumentations library, applied to test set only):

1. **Rain:** Synthetic droplets, reduced contrast
2. **Fog:** Atmospheric scattering (fog_coef=0.5)
3. **Low Brightness:** 30% brightness reduction

Evaluation Metrics:

- **Degradation Score:** Accuracy_corrupted / Accuracy_clean
- **Confusion Matrices:** 5×5 matrices per corruption type
- **Vulnerability Index:** Per-class susceptibility ranking

Timeline

Week	Milestone	Deliverables
1	Model Training	Trained ResNet-50 (>90% clean accuracy), dataset prepared
2	Corruption & Evaluation	3 corrupted test sets generated, all accuracy scores computed

Week	Milestone	Deliverables
3	Analysis	Confusion matrices, degradation scores, vulnerability rankings, visualizations
4	Documentation	6-page final report, code repository with README

Expected Results

Metric	Expected Range	Hypothesis
Clean accuracy	92-98%	Baseline performance
Rain degradation	below 0.85	Moderate impact
Fog degradation	**below 0.75**	**Most damaging** (loss of high-frequency details)
Brightness degradation	below 0.90	Least impact

Key Prediction: Speed limit signs (text-dependent) will be more vulnerable than Stop/Yield (shape-dependent).

Deliverables

✅ **Trained ResNet-50 model** with >85% clean accuracy ✅ **Degradation benchmarks** for 3 weather conditions ✅ **Confusion matrices** identifying common failure modes ✅ **Final report (6-8 pages)** with visualizations and actionable insights ✅ **Code repository** with training/evaluation scripts and README

Team Responsibilities

Member	Role
TBD	Evaluation pipeline, confusion matrix analysis
TBD	Corruption implementation, visualization
TBD	Augmentation testing, degradation metrics
TBD	ResNet-50 training, lead report writer

Resources

- **GPU:** Google Colab (free tier or Pro)
- **Software:** PyTorch 2.8+, Albumentations, matplotlib
- **Storage:** ~ 5 GB (GTSRB/LISA dataset + corrupted variants)

Success Criteria

Minimum: ResNet-50 trained (>85%), degradation scores for 3 corruptions, confusion matrices, 6-page report **Ideal:** >92% clean accuracy, clear vulnerability patterns identified, publication-quality analysis

References

1. Stallkamp et al. (2012). Man vs. computer: Benchmarking ML for traffic sign recognition. *Neural Networks*, 32, 323-332.
2. Hendrycks & Dietterich (2019). Benchmarking neural network robustness to common corruptions. *ICLR*.
3. Qiao, X. (2023). Traffic sign recognition based on CNN. *Procedia Computer Science*, 220, 107-114.